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RESEARCH ARTICLE-CIVIL ENGINEERING

Exploring the Sustainability Benefits of Digital Twin Technology in Achieving Resilient Smart Cities During Strong Earthquake Events

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Abstract
In recent years, the increasing frequency and intensity of earthquakes have highlighted the need for resilient urban planning, particularly in the context of smart cities. Despite advancements in technology and infrastructure, there remains a significant gap in understanding and effectively adopting digital twin technology for enhancing urban resilience against such seismic events. This study aims to conduct a critical assessment of using digital twin technology for seismic resilience in smart cities and to perform a systematic sustainability analysis to understand how digital twin technology can be utilized to build resilient smart cities that can better withstand and adapt to strong earthquake events. The research is motivated by the critical need to enhance urban earthquake resilience and the potential of digital twin technology as a transformative tool in urban planning and disaster management. By exploring the implementation of a digital twin model and performing a strengths, weaknesses, opportunities, and threats analysis, the study provides comprehensive insights into the dynamics of urban areas under seismic stress. It leverages real-time data and advanced simulation techniques to assess, predict, and strategize responses to earthquake scenarios, enhancing urban resilience and sustainability. In addition, this research demonstrates how digital twin technology can help in achieving several United Nations sustainable development goals, particularly those related to sustainable cities and communities, industry, innovation, and infrastructure, as well as climate action, which are represented in goals 9, 11, and 13. Ultimately, the study findings highlight the importance of integrating advanced technological solutions, such as digital twins, in urban development strategies to build smarter, more resilient cities capable of withstanding future seismic challenges.

Keywords Digital twin technology · Urban earthquake resilience · Smart cities · Sustainable development goals · Seismic disaster management

1 Introduction

The intensity of seismic events worldwide has brought to the forefront the urgent need for resilient urban planning [1, 2]. Smart cities hold significant promise in this regard, particularly through the application of digital twin technology, thanks to their advanced technological infrastructure [3–7]. Digital twins, as virtual replicas of physical entities, can facilitate real-time monitoring and predictive analysis, enhancing urban resilience against earthquakes and other disasters. Recently, the concept of digital twins has gained considerable attention in various fields, including urban planning and disaster management. For instance, Alibrandi [8] and Ariyachandra and Wedawatta [9] both emphasized the importance of digital twins in developing risk-informed models for urban infrastructure and their role in disaster risk management within smart cities. They highlighted how digital twins aid in understanding and mitigating urban risks. Sustainability is another critical aspect in the built environment that is being investigated a lot nowadays [10–16]. In this regard, Barni et al. [17] explored the use of digital twins in assessing and optimizing the sustainability performances of urban developments, indicating their crucial

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role in sustainable urbanization. The versatility, characteristics, and applications of digital twin applications in various urban scenarios was further detailed by Barricelli et al. [18]. Bhandal et al. [19] discussed the relevance of digital twins in operations and supply chain management, showing their cross-sectoral significance, including in urban resilience and disaster management. Boschert et al. [20] introduced the concept of next-generation digital twins, underlining the progressive evolution of this technology and its growing importance in smart city frameworks. Botín-Sanabria et al. [21] reviewed the challenges and diverse applications of digital twins in urban settings. Chaudhary et al. [22] examined various aspects of digital twin technology, providing insights into its applications and limitations. This study specifically aims to evaluate the sustainability benefits of digital twin technology in enhancing urban resilience against seismic events, addressing a significant gap in the current literature. Evangeline [23] and Fan et al. [24] discussed the integration of digital twins in smart manufacturing and social sensing during built environment disruptions, laying a foundation for their application in earthquake resilience. Fan et al. [25] envisioned the disaster city digital twin, integrating artificial and human intelligence for disaster management. Multiple similar studies on using artificial intelligence to achieve resilience have been conducted in the past years [26–33]. Other studies have discussed various other simulation methods and technologies as well for this purpose [34–41]. On the other hand, Farsi et al. [42], Ford and Wolf [43], and Fuller et al. [44] provided broader perspectives on the potential of digital twins in enhancing urban intelligence, sustainability, and resilience. Hayat [45] and He and Bai [46] highlighted the role of digital twins in global urban development and sustainable intelligent manufacturing. He et al. [47] discussed the data-driven aspects of digital twins for optimized control, adaptable for urban resilience strategies. Jones et al. [48] offered a foundational understanding of digital twins, setting the stage for their application in earthquake resilience. Kaewunruen & Xu [49], Kaewunruen et al. [50–52], and Kaewunruen and Lian [53] extensively explored the application of digital twins in urban infrastructure, highlighting their practical implications in assessing and maintaining infrastructure sustainability and vulnerability. Khajavi et al. [54] provided insights into the benefits and boundaries of digital twins for buildings, relevant in earthquake resilience. Madni et al. [55] utilized digital twins in model-based systems engineering, adaptable for urban planning and disaster management. Malykhina and Tarkhov [56] and Mashaly [57] discussed the technological advancements and requirements for implementing digital twins in smart cities. Onan Demirel et al. [58] proposed digital twin-driven human-centered design frameworks for sustainability, critical in designing earthquake resilient environments. Park and Yang [59] demonstrated a GIS-enabled digital twin system for sustainable carbon emissions evaluation. Pirlone et al. [60] emphasized the need for advanced technologies such as digital twins in resilient city strategies. Qi et al. [61] highlighted enabling technologies and tools for digital twins, providing insights into their technical implementation in urban scenarios. Samarakkody et al. [62] analyzed technological innovations for disaster resilience in smart cities, reinforcing the role of digital twins. Shah et al. [63] explored the integration of IoT and big data analytics in creating resilient smart cities, providing a framework that can be combined with digital twin technology. Tagliabue et al. [64] used digital twins for sustainability assessment in educational buildings, showing their application in urban infrastructure evaluation. Tahmasebinia et al. [65] explored the benefits and limitations of digital twins in building energy management, extendable to earthquake resilience contexts. Tao et al. [66, 67] provided insights into digital twin modeling and its applications in industry, informing its implementation in urban earthquake resilience. VanDerHorn and Mahadevan [68] characterized digital twin technology and proposed a framework for its use in decision support systems. Wang [69] discussed the relevance of digital twins in Industry 4.0, applicable to smart urban planning. Yu and He [70], and Yu et al. [71] investigated digital twin-driven intelligence for disaster prevention and energy management in urban areas. Finally, Zhang et al. [72] reviewed the role of digital twins in enhancing sustainability in energy-positive districts. Zhu et al. [73, 74] compared smart and resilient cities, providing insights into their differences and connections. These studies collectively illustrate the significant potential of digital twin technology in fostering resilient, sustainable, and intelligent urban environments, particularly in the face of seismic and other disaster events. Accordingly, while advancements in technology and infrastructure have been significant, their application in smart city planning for earthquake resilience remains inadequately understood and implemented. This gap is particularly concerning given the rising frequency and intensity of earthquakes globally, which pose a significant threat to urban areas. Therefore, this study aims to conduct a critical assessment of using digital twin technology for seismic resilience in smart cities and to perform a systematic sustainability analysis to examine how digital twin technology can foster resilient smart cities that are better equipped to withstand and adapt to seismic events. The importance of this study is manifold. Firstly, it responds to the urgent need for enhanced urban earthquake resilience. Earthquakes can have devastating impacts on cities, affecting lives, economies, and the environment. Therefore, improving resilience against such events is crucial. Secondly, the study positions digital twin technology as a transformative tool in urban planning and disaster management. By employing this technology, the study explores new ways to integrate real-time data and advanced simulation techniques for more effective response

strategies to earthquake scenarios. In addition, the study's importance extends to its potential contribution to achieving several United Nations sustainable development goals (SDGs). By enhancing urban resilience and sustainability, the findings can directly impact goals related to sustainable cities and communities, industry, innovation, infrastructure, and climate action (Goals 9, 11, and 13). This aligns the study with broader global efforts toward sustainable development. Therefore, the study is significant for its potential to fill a crucial gap in urban resilience planning, particularly in the context of seismic events. Its focus on digital twin technology not only provides a novel approach to disaster management but also aligns with international sustainability objectives, highlighting its global relevance and urgency.

2 Digital Twin Technology in Urban Planning

Digital twin technology is an innovative concept in urban planning and development, which is being increasingly recognized for its potential to enhance urban resilience, especially in scenarios such as earthquake preparedness and response [75–77]. It refers to the creation of a highly detailed digital replica of a physical entity, system, or process. This concept has emerged as a revolutionary tool in urban planning, providing a virtual model that mirrors the real-world scenario of a city or urban area. In this regard, digital twins facilitate dynamic, real-time monitoring and simulation of urban environments with the help of advanced data analytics, IoT, and computational modeling [78–82]. This enables stakeholders to observe, analyze, and predict the behavior and evolution of urban systems under various scenarios [83, 84]. Charitonidou [85] further elaborates that in the context of urban planning, digital twins are not just replicative models but adaptive systems that integrate and synthesize complex urban data, enhancing decision-making in a data-driven society. The key components of a digital twin in urban planning include a real-world physical model (the city or urban area), its digital counterpart (the twin), a continuous data exchange mechanism ensuring real-time updates, and analytics tools for interpretation and decision-making. These components work collaboratively to provide a comprehensive, interactive, and up-to-date model of urban spaces [86, 87]. On the other hand, the layers of a of a digital twin for a smart city, Table 1, includes terrain, buildings, infrastructure, mobility, and virtual smart city. In general, digital twin technology provides a broad spectrum of theoretical applications for enhancing urban resilience. For instance, it can be utilized to simulate natural disasters such as earthquakes. This capability is invaluable for urban planners and policymakers, as it allows them to anticipate the impact of such events on city infrastructure, public safety, and resource allocation. In addition, digital twins can help identify vulnerabilities in urban infrastructure, predict the repercussions of seismic activities, and devise optimized disaster response and mitigation strategies. These theoretical applications are pivotal in preparing cities to effectively manage and recover from earthquake events [88, 89]. In addition, they facilitate informed decision-making and the development of proactive strategies for disaster management. For example, through simulation, digital twins can predict the effects of an earthquake on critical infrastructure, such as bridges, buildings, and utility networks, identifying potential risks and allowing for preventive measures. Post-earthquake, digital twins can assist in efficient resource management, rapid damage assessment, and streamlined recovery operations. In addition, by integrating real-time data and citizen feedback, digital twins can enhance community engagement and preparedness, fostering a more resilient urban populace [79, 90]. In addition, the use of digital twin technology in urban planning for earthquake scenarios supports sustainable city development. By enabling a better understanding and management of urban systems, digital twins contribute to the optimization of resource use, reduction in environmental impacts, and enhancement of overall urban sustainability. They allow for the testing of various urban development scenarios, ensuring that city growth aligns with sustainability goals and resilience to natural disasters [83, 84]. Figure 1 illustrates the key components.

3 Smart Cities and Earthquake Resilience

The concept of smart cities has been a significant focus in urban development and planning, particularly in the context of building resilience against natural disasters such as earthquakes [91, 92]. This section explores the concept and characteristics of smart cities, the challenges they face during seismic events, and the role of technology in enhancing earthquake resilience, as detailed in Fig. 1. Smart cities are urban areas that use different types of electronic methods and sensors to collect data, which is then utilized to manage assets, resources, and services efficiently [93]. These data are collected from citizens, buildings, devices, and assets and then processed and analyzed to monitor and manage traffic and transportation systems, power plants, utilities, water supply networks, waste, crime detection, information systems, schools, libraries, hospitals, and other community services. The hallmark of a smart city is its ability to use technology to improve efficiency and enhance the quality of life for its inhabitants [45]. Earthquake events pose significant challenges to cities, primarily due to the immediate impact on infrastructure and the subsequent disruption of services [94–98]. The primary challenge is ensuring the resilience

Table 1 Layers of digital twin technology in cities and urban planning

Layer	Description	Key components
Layer 0: Terrain	This layer represents the natural and geographical base of the city	Digital elevation models (DEM) Soil composition Water bodies Vegetation Geological features
Layer 1: Buildings	Focuses on the physical structures within the city	Architectural models Building materials Structural health Energy consumption patterns Occupancy patterns
Layer 2: Infrastructure	Covers the essential facilities and systems serving the city	Utilities (water, electricity, gas) Waste management systems Telecommunications networks Public amenities (parks, schools)
Layer 3: Mobility	Deals with the movement of people and goods throughout the city	Public transport networks (buses, trains) Road networks Traffic patterns Pedestrian flow Vehicle-to-infrastructure communication
Layer 4: Virtual smart city	Integrates data and digital services to enhance city management and services	IoT sensors and devices Real-time data analytics Urban dashboards Smart public services (lighting, parking) Emergency response systems

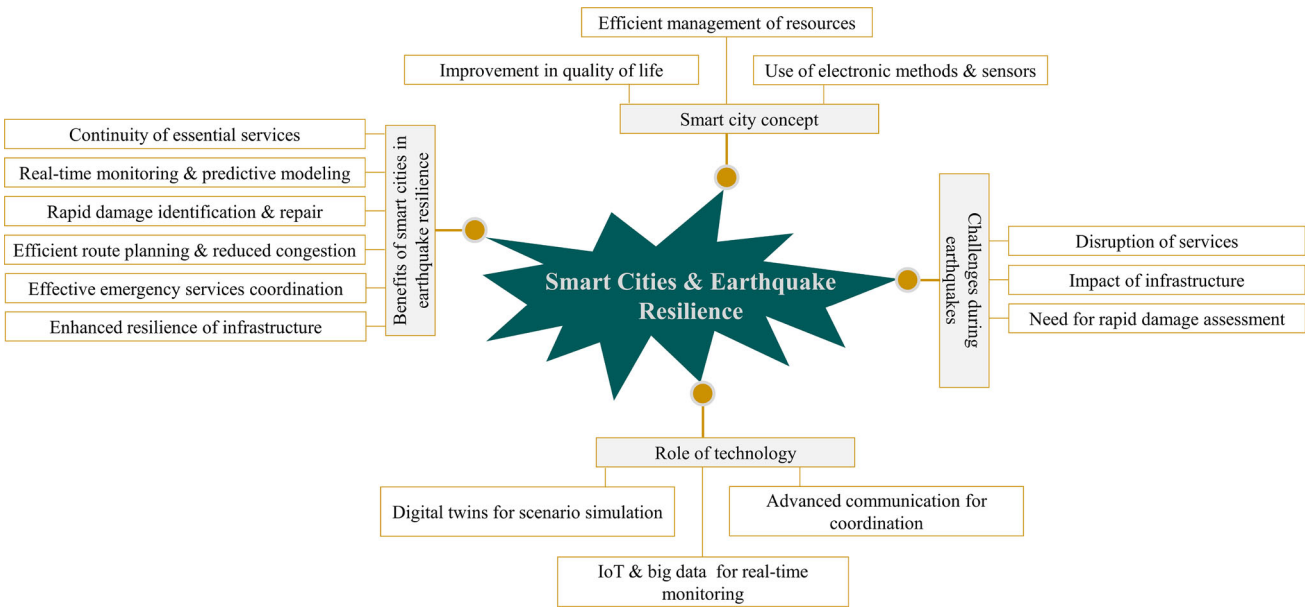


Fig. 1 Concepts, challenges, technological roles, and benefits of smart cities in enhancing earthquake resilience

of critical infrastructure, such as power grids, transportation networks, and communication systems, against seismic activities. In addition, the need for efficient emergency response mechanisms and rapid damage assessment is crucial in the aftermath of an earthquake. The challenge extends to the recovery phase, where smart cities need to rebuild and repair infrastructure while also addressing the social and economic impacts of the disaster [73]. Technology plays a pivotal role in enhancing the resilience of smart cities to earthquakes [99]. The integration of the Internet of Things (IoT), big

data analytics, and advanced communication technologies enables real-time monitoring and rapid response during seismic events [63, 100]. Technologies such as sensors and IoT devices can detect early warning signs of earthquakes, allowing for timely evacuation and preparedness measures [101, 102]. Big data analytics aids in processing large volumes of data to predict the impact of earthquakes and plan effective response strategies [103–106]. In addition, advanced communication technologies ensure continuous information flow, which is critical for coordination among emergency services and the public during and after an earthquake [63]. The integration of digital twin technology in smart city frameworks significantly enhances earthquake resilience. Digital twins, as virtual replicas of physical city environments, allow for the simulation and analysis of seismic impacts on urban infrastructure and populations. This technology enables city planners and decision-makers to visualize and test various earthquake scenarios, leading to better-prepared and more resilient urban systems.

Digital twins can also be used for training purposes, allowing emergency responders to simulate different scenarios and develop effective response strategies (Zhu et al., 2020). Table 2 summarizes the main applications of smart cities in disaster management.

4 Digital Twin Technology for Seismic Risk Resilience in Smart Cities

4.1 Implementation of Digital Twin Technology in Smart Cities

The integration of digital twin technology in smart cities presents a strategic approach to enhance seismic risk resilience. This involves a multi-phased implementation, Fig. 2, with variety of factors and measures, Table 3, to establish a robust technical infrastructure and engage stakeholders effectively. In general, the integration of digital twin technology in urban planning for seismic risk resilience involves several phases. Initially, the focus is on conceptualizing and defining the scope of the digital twin, which requires a thorough understanding of the city's infrastructure, dynamics, and seismic risk factors [55, 69]. This phase involves the collection and analysis of extensive urban data, using technologies such as IoT, GIS, and big data analytics [44, 61]. Subsequent phases involve the development and testing of the digital twin model to ensure its accuracy and reliability in simulating seismic impacts. This phase is critical for understanding how different parts of the urban environment interact and respond to seismic events [22, 68]. The final phases focus on the implementation of the digital twin in real-world scenarios, continuous refinement, and the use of the model for decision-making in urban planning and emergency response [47, 107].

The technical infrastructure for a digital twin in urban planning for seismic resilience comprises several key components. At its core, the infrastructure requires high-performance computing systems capable of processing and analyzing large datasets in real-time [56, 66]. This is complemented by an extensive network of sensors and IoT devices deployed throughout the city to collect real-time data on various parameters such as structural integrity, traffic movement, and environmental conditions [42, 57].The system architecture of the digital twin must be robust, scalable, and interoperable, allowing for seamless integration with existing urban management systems and databases [18, 23]. In addition, it requires advanced modeling and simulation tools capable of creating accurate and dynamic representations of the urban environment [20, 48].

Stakeholder engagement is vital for the successful implementation of digital twin technology in urban planning. This involves collaboration with various entities, Table 4, such as government agencies, urban planners, emergency responders, technology experts, and the community at large [67, 108]. Engaging stakeholders in the early stages of development helps in understanding their needs, expectations, and concerns, ensuring that the digital twin model is relevant and effective in addressing the specific challenges of seismic resilience [19, 83]. In addition, it helps in using diverse perspectives and expertise, which is essential for the continuous improvement and evolution of the digital twin model [79, 90]. Community involvement, in particular, is crucial as it ensures that the digital twin is aligned with the needs of the residents and is used effectively in enhancing their preparedness and response to seismic events [84, 85]. Collaborative approaches also involve knowledge sharing and capacity building among stakeholders, enabling them to effectively use and benefit from the digital twin technology [88, 89]. Through strategic implementation, robust technical infrastructure, and effective stakeholder engagement, digital twin technology can significantly enhance the seismic risk resilience of smart cities. It provides urban planners and decision-makers with a powerful tool for simulating, predicting, and managing the impacts of seismic events, contributing to safer, more resilient urban environments [80, 86]. The use of digital twins in smart cities represents a significant advancement in urban planning, providing a new paradigm for managing complex urban systems and challenges [87].

4.2 SWOT Analysis of Digital Twins in Urban Seismic Resilience

The deployment of digital twin technology in urban seismic resilience requires a strengths, weaknesses, opportunities, and threats (SWOT) analysis. This examination helps to

Table 2 Smart city applications of in disaster management

Smart city aspect	Technology/method	Benefits of earthquake resilience	Implementation challenges	Potential for scale-up	Impact measurement
Data collection and analysis	IoT sensors, big data analytics	Real-time monitoring, Predictive modeling for risk assessment	Data privacy concerns, Integration with existing systems	High, applicable to various urban scales	Reduced response times, Accuracy of predictions
Traffic and transportation management	Smart traffic systems, GPS data analysis	Efficient route planning during evacuations, Reduced congestion	Need for widespread sensor deployment, Reliance on real-time data	Moderate, varies with city size and infrastructure	Decrease in evacuation time, Traffic flow improvement
Utility and resource management	Smart grids, Water supply network sensors	Continuity of essential services, Rapid damage identification and repair	High initial investment, Technical expertise	High, adaptable to different urban environments	Service uptime, Speed of damage recovery
Emergency services coordination	Advanced communication technologies	Effective communication among responders, Timely public alerts	Dependence on network infrastructure, Interoperability	High, essential for all city scales	Efficiency of emergency response, Public safety levels
Building and infrastructure safety	Seismic retrofitting, Structural health monitoring	Enhanced resilience, Early detection of structural weaknesses	High costs, Disruption during retrofitting	High in earthquake-prone areas	Reduction in building collapse incidents, Enhanced public safety
Health and public services management	e-Health systems, Remote monitoring	Continuity of healthcare, Rapid medical emergency response	Data security, Reliance on digital infrastructure	High, critical for urban health management	Access to healthcare, Response time in emergencies
Digital twin technology	Virtual replicas, Simulation software	Simulation of earthquake impacts, Preparedness training for responders	Technical complexity, High computational resources	Moderate, depending on city's digital infrastructure	Accuracy in scenario simulation, Effectiveness of training
Social and economic recovery	Big data analytics, Community engagement platforms	Efficient recovery planning, Assessing social and economic impacts	Need for comprehensive data, Community participation	High, scalable to various community sizes	Speed of recovery, Economic stability post-disaster
Crime detection and security	Surveillance systems, AI-based analysis	Public order maintenance, Prevention of post-disaster crimes	Privacy concerns, Reliance on continuous power supply	Moderate, varies with urban safety needs	Reduction in crime rates, Public perception of safety
Environmental monitoring	Environmental sensors, Data analysis	Environmental impact assessment, Planning for environmental recovery	Sensor accuracy, Data interpretation	High, applicable to diverse ecosystems	Environmental quality indices, Effective recovery actions

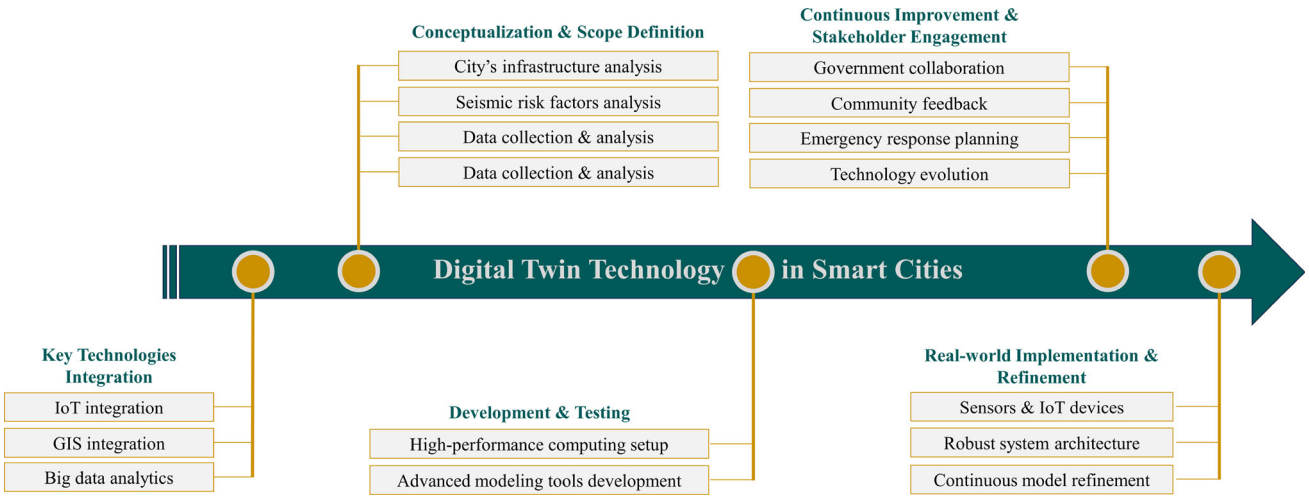


Fig. 2 A flowchart for the implementation of digital twin technology in smart cities

Table 3 Factors and measures for the implementation of digital twin technology in smart cities

Phase	Factors	Key technologies	Implementation measures	Expected outcomes
Conceptualization & scope definition	City's infrastructure analysis Seismic risk factors	IoT GIS Big Data Analytics	Data collection and analysis Understanding urban dynamics	Comprehensive scope for digital twin
Development & testing	Urban environment interaction Seismic impact simulation	High-performance computing Advanced modeling tools	Development of accurate digital twin model Reliability and accuracy testing	Effective seismic impact simulation
Real-world implementation & refinement	Decision-making in urban planning Emergency response enhancement	Sensors and IoT devices Robust system architecture	Deployment in real scenarios Continuous model refinement	Improved urban planning and emergency response

understand the impact of digital twins in enhancing urban resilience to seismic risks. Table 5 summarizes the results of the SWOT analysis in this study.

The primary strength of digital twin technology in urban seismic resilience lies in its ability to enhance real-time response and predictive capabilities. Digital twins facilitate real-time monitoring and simulation of urban environments, enabling city planners and emergency services to respond more effectively to seismic events [55, 107]. In addition, risk-informed digital twin frameworks for buildings and infrastructures can significantly reduce vulnerabilities by providing continuous feedback on structural health and potential hazards [8, 21]. These capabilities are further reinforced by integrating advanced data analytics and situational awareness approaches, as illustrated in disaster city digital twin frameworks [24, 109]. The predictive capabilities of digital twins, supported by data-driven models, allow for the anticipation of seismic impacts and the preparation of more effective mitigation strategies [44, 61]. This technology, as characterized by Chaudhary et al. [22] and VanDerHorn & Mahadevan [68], provides a dynamic and adaptable platform for optimizing urban resilience measures.

Despite these strengths, digital twin technology faces several weaknesses, primarily in terms of technical, financial, and operational limitations. The complexity of developing and maintaining accurate and comprehensive digital twins poses significant technical challenges, especially at larger scales [47, 69, 80]. Financial constraints, particularly the initial investment and ongoing maintenance costs, can be prohibitive for many cities [56, 66]. Operational challenges include the need for skilled personnel and the integration of digital twins with existing urban management systems [42, 57]. In addition, issues related to data availability and quality can hamper the successful implementation of digital twins in disaster-prone areas [86].

The deployment of digital twin technology in urban seismic resilience opens numerous opportunities. It fosters innovation in urban planning and management, particularly in using advanced technologies for risk assessment and response [9, 18, 23]. Digital twins can also contribute to

Table 4 Entities involved in digital twin implementation and their roles

Entity	Role	Contributions	Importance in implementation
Urban planners	Planning and design integration	Incorporate digital twin technology into urban design Utilize insights for seismic risk mitigation strategies	Bridge the gap between technology and practical urban development Ensure the digital twin serves urban resilience
Technology experts	Technical development and support	Architect and maintain the digital twin infrastructure Provide expertise in emerging technologies	Drive the development of a reliable and scalable digital twin framework Innovate and solve technical challenges
Data scientists	Analytics and modeling	Analyze extensive urban data for actionable insights Model urban dynamics and seismic impact simulations	Enhance predictive accuracy of digital twins Provide data-driven decisions for urban planning
IT team	Infrastructure management	Set up and maintain high-performance computing systems Ensure system security and data integrity	Secure the backbone for digital twin operations Guarantee data reliability and system integrity
Software developers	Software engineering	Develop custom software solutions for digital twin integration Ensure interoperability and real-time data processing	Create the functional capabilities of digital twins Enable seamless interaction with urban data sources
Field engineers	Field data collection and system validation	Deploy sensor networks and IoT devices for real-time data collection Validate digital twin models against real-world conditions	Ensure the digital twin's data accuracy and relevance Bridge the gap between virtual models and physical reality
Government agencies	Policy and financial support	Establish legal and financial frameworks Align digital twin initiatives with urban planning and safety goals	Provide essential legislative and financial backing Facilitate interdepartmental coordination
Community	Feedback provision and adaptation	Supply insights and feedback based on local knowledge Adapt digital twin usage to community needs	Align digital twin projects with the needs of residents Foster community engagement and support
Emergency responders	Operational utilization	Apply digital twin for real-time emergency planning and response	Enhance emergency preparedness and response efficiency Use digital twin for situational awareness

sustainability by optimizing resource use and reducing the environmental impact of urban responses to seismic events [20, 48, 53]. In addition, the concept of "urban scale digital twins" introduces new paradigms for data-driven urban governance, enabling more transparent and participatory decision-making processes [67, 85, 108]. This data-centric approach also aligns with the increasing emphasis on smart cities, which strive to enhance disaster resilience through

Table 5 SWOT analysis of digital twins in urban seismic resilience

Strengths	Weaknesses	Opportunities	Threats
Real-time response enhancement	Technical challenges in development and maintenance	Innovation in urban planning and management	Security risks (cyber-physical systems vulnerability)
Predictive capabilities for seismic impacts	Financial constraints (initial investment and maintenance costs)	Sustainability and optimized resource use	Data privacy concerns
Dynamic and adaptable platform for resilience measures	Operational challenges (need for skilled personnel, integration issues)	Smart governance through transparent, participatory decision-making	Dependence on technology and resilience to technological failures

integrated platforms that combine physical and digital layers [43, 63].

However, the use of digital twins in urban seismic resilience is not without threats. Security risks associated with the cyber-physical systems of digital twins are a significant concern, as they can be targets for cyberattacks that disrupt urban responses to seismic events [19]. Data privacy is another critical issue, given the sensitive nature of the data collected and used in digital twins [83]. In addition, the increasing dependence on technology raises concerns about the resilience of cities in the event of technological failures or disruptions [84]. As a result, ensuring the robustness, reliability, and security of these systems is paramount for their successful adoption in urban seismic resilience [70].

Finally, this SWOT analysis highlights both the potential and the challenges of using digital twins in urban seismic resilience. Continued progress in addressing these weaknesses and threats through robust cybersecurity measures, integrated data governance frameworks, and adequate resource allocation will be crucial for getting the full benefits of digital twin technology in seismic risk management.

4.3 Sustainability Benefits and Impact Analysis

The integration of digital twin technology in urban planning, particularly in enhancing seismic resilience, represents a sustainable approach to urban development [8]. This technology not only prepares cities for seismic challenges but also commits to sustainable growth and development, effectively bridging the gap between immediate disaster response and long-term environmental stewardship [24]. Table 6 provides an impact analysis of digital twin technology on urban sustainability and resilience with respect to the United Nations' SGDs. For instance, in seismic risk resilience of smart cities, digital twins play a crucial role in fostering sustainability benefits through enhanced resource efficiency and management [43]. Digital twins serve an essential role in managing water and utilities, which is especially critical during seismic events where infrastructure is at risk [50]. Their real-time monitoring capabilities ensure that these services are effectively managed and restored sustainably, aligning with the minimized environmental impact and raising sustainable urban ecosystems [65]. In post-earthquake scenarios, where waste management is a significant challenge, digital twins guide waste collection and disposal, reducing environmental hazards and reinforcing sustainable urban living [49]. Emergency resource allocation is another area where digital twins demonstrate their effectiveness [109]. In the chaos following an earthquake, they ensure the quick and efficient distribution of vital resources, optimizing their use and minimizing wastage [47]. This targeted allocation is crucial for immediate relief and exemplifies sustainable emergency response practices. In addition, digital twin technology enhances the sustainability of urban infrastructure [48]. It provides insights into the condition and resilience of infrastructure, guiding maintenance and retrofitting to withstand seismic activities [52]. This not only addresses immediate seismic risks but also ensures long-term sustainability by extending the life of existing structures and raising the development of new, sustainable, and resilient ones [17] Therefore, digital twin technology aligns perfectly with SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). In this regard, digital twins advance SDG 9 by contributing to resilient infrastructure development, where they foster innovation at the heart of modern resilient urban planning [67]. In the context of SDG 11, digital twins ensure the sustainability of essential services, contributing to the livability and resilience of cities post-earthquakes [61]. Their role in emergency resource allocation also reflects a commitment to making cities safer, more resilient, and sustainable, encompassing both physical and social dimensions [59]. Regarding SDG 13, digital twins are instrumental in combatting climate change [63]. By optimizing resource use and reducing waste, they aid in lowering greenhouse gas emissions and play a vital role in adapting to and mitigating the impacts of climate change [72]. This is especially pertinent in urban areas, where balancing growth and environmental impact is crucial.

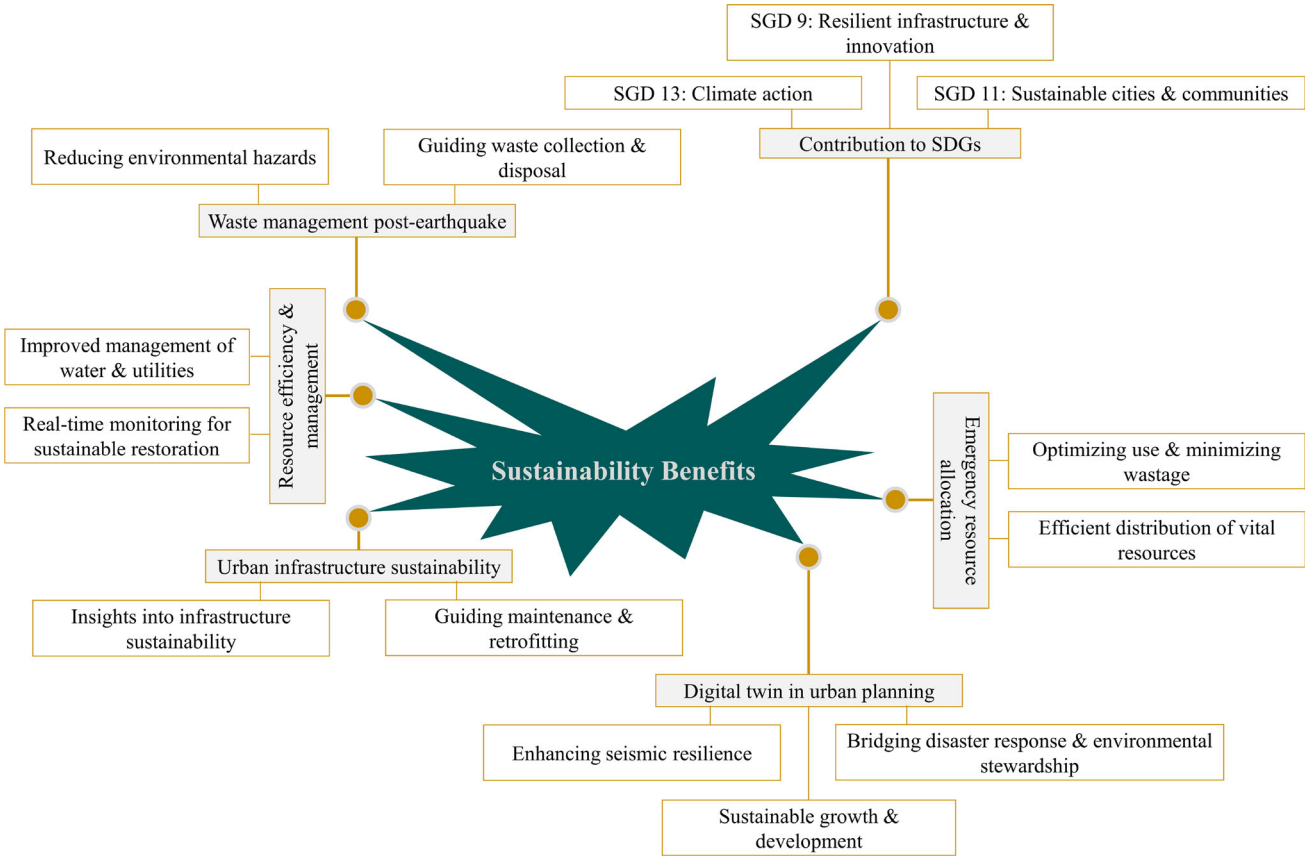


Fig. 3 Sustainability benefits of digital twin technology in achieving resilient smart cities during strong earthquake events

Accordingly, the integration of digital twin technology in urban planning and management, particularly for seismic resilience, considerably benefits the achievement of key sustainable development goals, as shown in Fig. 3. By improving resource efficiency and management, digital twins contribute to building resilient infrastructure, making cities safer and more sustainable, and acting against climate change [62]. This technology demonstrates how innovative solutions can be harnessed to address pressing challenges, paving the way for a more resilient and sustainable future in urban environments [21].

5 Discussions and Future Implications

The discussion regarding the use of digital twin technology in urban planning, particularly in the context of seismic events, opens up a vast landscape of future implications and possibilities. The research presented here highlights the transformative potential of digital twins in enhancing urban resilience, fostering sustainable development, and addressing the urgent challenges posed by increasing seismic activities globally. The application of digital twins in smart cities for earthquake resilience represents a paradigm shift in how urban environments prepare for and respond to natural disasters. The ability of digital twins to simulate real-world scenarios in real-time provides city planners and emergency responders with an unprecedented level of insight and preparedness. This capability is not only crucial for immediate disaster response but also for long-term urban planning and sustainability. Future research and development in this field are likely to focus on further enhancing the accuracy and predictive capabilities of digital twins, integrating them more seamlessly with other urban management systems, and expanding their application to a broader range of urban challenges. However, some challenges need to be addressed as digital twin technology becomes more integral to urban planning. These include addressing the technical and financial barriers to implementation, ensuring the security and privacy of the data involved, and fostering the necessary skills and knowledge among urban planners and decision-makers. In addition, the increasing reliance on technology for urban resilience raises concerns about the potential consequences of technological failures or disruptions. Future efforts must, therefore, prioritize the development of robust, secure, and resilient digital twin systems. The sustainability implications of digital twin technology are particularly significant. By enabling more efficient resource management,

Table 6 Impact analysis of digital twin technology on urban sustainability and resilience with respect to the United Nations' SGDs

Key focus area	Specific benefits	Related SDGs
Seismic resilience in urban planning	Enhancing seismic resilience, preparing cities for seismic challenges, bridging disaster response and environmental stewardship	SDG 9, SDG 11, SDG 13
Resource efficiency and management	Improved management of water and utilities, real-time monitoring for sustainable restoration	SDG 11, SDG 13
Waste management in post-earthquake scenarios	Guiding waste collection and disposal, reducing environmental hazards	SDG 11
Emergency resource allocation	Efficient distribution of vital resources, optimizing use, minimizing wastage	SDG 11, SDG 13
Urban infrastructure sustainability	Insights into infrastructure condition and resilience, guiding maintenance and retrofitting	SDG 9, SDG 11
Contribution to resilient infrastructure development	Fostering innovation in urban planning, enhancing infrastructure resilience	SDG 9
Enhancing livability and resilience of cities	Ensuring sustainability of essential services, improving city resilience post-earthquakes	SDG 11
Combatting climate change	Optimizing resource use, reducing waste, lowering greenhouse gas emissions	SDG 13
Overall impact on sustainable development goals	Building resilient infrastructure, making cities safer and sustainable, acting against climate change	SDG 9, SDG 11, SDG 13

reducing environmental impacts, and aiding in sustainable recovery efforts post-disasters, digital twins align closely with the United Nations Sustainable Development Goals. This alignment opens up new avenues for cities to become more resilient, not just to seismic events but also to the broader challenges of sustainable urban development. Looking ahead, the integration of digital twin technology in urban planning provides a promising pathway to more resilient, sustainable, and intelligent cities. It presents an opportunity to rethink urban resilience in the face of growing seismic risks and to harness the power of technology in service of a more sustainable future. The continued evolution of digital twins is likely to play a critical role in shaping the cities of tomorrow, making them better equipped to withstand and adapt to the challenges posed by natural disasters and climate change.

6 Conclusion

This study aimed to conduct a critical assessment of using digital twin technology for seismic resilience in smart cities and to perform a systematic sustainability analysis to evaluate how digital twin technology can enhance the resilience of smart cities, particularly in the face of strong earthquake events. Addressing the significant gap in the literature, the study focused on the integrated use of digital twins for enhancing earthquake resilience in smart cities. This relatively unexplored area holds critical potential for urban planning and disaster management. The major conclusions of this study are as follows:

1. Digital twin technology significantly improves real-time monitoring and predictive analysis in urban settings, enabling cities to anticipate and respond more effectively to earthquake events.
2. Despite its potential, the integration of digital twin technology in urban systems faces technical, financial, and operational barriers. However, with strategic planning and stakeholder engagement, these challenges can be effectively addressed.
3. The use of digital twins aligns well with various SDGs, particularly in fostering sustainable cities and communities, responsible consumption, and climate action.
4. Effective implementation of digital twin technology requires supportive policy frameworks and regulatory environments. Public–private partnerships emerge as crucial in driving innovation and investment in this field.
5. The future of digital twin technology in urban resilience looks promising, with potential enhancements focusing on artificial intelligence integration, advanced risk assessment, and increased focus on social dynamics.

This study provides valuable insights but it is not without limitations. The primary constraint lies in the rapidly evolving nature of digital twin technology, which can lead to changes in its applications and capabilities post-study. In addition, the focus on urban earthquake resilience does not include the full spectrum of digital twin applications in other

types of urban challenges. Future studies should explore the evolving nature of digital twin technology, particularly in the context of different types of natural disasters and urban challenges. In addition, they should investigate the long-term economic impacts of implementing digital twin technology in urban resilience strategies. In addition, there is a need for more empirical studies to validate the theoretical frameworks and models discussed, ensuring that digital twin technology's potential is fully realized in practical, real-world scenarios.

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Data Availability Some or all data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest to report regarding the present study.

References

1. Zhao, T.; Wang, J.; Sun, L.: Crowd dynamic-based model on the city-wide emergency transfer under catastrophic earthquakes. Int. J. Disaster Risk Reduct. **93**, 103777 (2023)
2. Hu, J.; Wen, W.; Zhai, C.; Pei, S.: A comprehensive review of resilience of urban metro systems: a perspective from earthquake engineering. Tunn. Undergr. Space Technol. **152**, 105920 (2024)
3. Piras, G.; Agostinelli, S.; Muzi, F.: Digital twin framework for built environment: a review of key enablers. Energies **17**(2), 436 (2024)
4. Gkontzis, A.F.; Kotsiantis, S.; Feretzakis, G.; Verykios, V.S.: Enhancing urban resilience: smart city data analyses, forecasts, and digital twin techniques at the neighborhood level. Future Internet **16**(2), 47 (2024)
5. Habib, A.; Alnaemi, A.; Habib, M.: Developing a framework for integrating blockchain technology into earthquake risk mitigation and disaster management strategies of smart cities. Smart Sustain. Built Environ. (2024). https://doi.org/10.1108/SASBE-12-2023-0376
6. Luo, J.; Liu, P.; Xu, W.; Zhao, T.; Biljecki, F.: A perception-powered urban digital twin to support human-centered urban planning and sustainable city development. Cities **156**, 105473 (2025)
7. Argota Sánchez-Vaquerizo, J.: Urban Digital Twins and metaverses towards city multiplicities: uniting or dividing urban experiences? Ethics Inf. Technol. **27**(1), 1–31 (2025)
8. Alibrandi, U.: Risk-informed digital twin of buildings and infrastructures for sustainable and resilient urban communities. ASCE-ASME J. Risk Uncertain. Eng. Syst. Part A: Civ. Eng. **8**(3), 04022032 (2022)
9. Ariyachandra, M.M.F.; Wedawatta, G.: Digital Twin Smart Cities for disaster risk management: a review of evolving concepts. Sustainability **15**(15), 11910 (2023)
10. Malkawi, A.B.; Habib, M.; Alzubi, Y.; Aladwan, J.: Engineering properties of lightweight geopolymer concrete using palm oil clinker aggregate. GEOMATE J. **18**(65), 132–139 (2020)
11. Al Houri, A.; Habib, A.; Elzokra, A.; Habib, M.: Tensile testing of soils: history, equipment and methodologies. Civ. Eng. J. **6**(3), 591–601 (2020)
12. Sepasgozar, S.M.: Differentiating digital twin from digital shadow: elucidating a paradigm shift to expedite a smart, sustainable built environment. Buildings **11**(4), 151 (2021)
13. Çetin, S.; De Wolf, C.; Bocken, N.: Circular digital built environment: an emerging framework. Sustainability **13**(11), 6348 (2021)
14. Khan, S.A.; Koç, M.; Al-Ghamdi, S.G.: Sustainability assessment, potentials and challenges of 3D printed concrete structures: a systematic review for built environmental applications. J. Clean. Prod. **303**, 127027 (2021)
15. Habib, M.: Quantifying topographic ruggedness using principal component analysis. Adv. Civ. Eng. **2021**(1), 3311912 (2021)
16. Habib, M.: Fit-for-purpose conformal mapping for sustainable land administration in war-ravaged Syria. Heliyon **8**(5), e09384 (2022)
17. Barni, A., Fontana, A., Menato, S., Sorlini, M., & Canetta, L.: Exploiting the digital twin in the assessment and optimization of sustainability performances. In 2018 International Conference on Intelligent Systems (IS), pp. 706–713. IEEE (2018)
18. Barricelli, B.R.; Casiraghi, E.; Fogli, D.: A survey on digital twin: Definitions, characteristics, applications, and design implications. IEEE Access **7**, 167653–167671 (2019)
19. Bhandal, R.; Meriton, R.; Kavanagh, R.E.; Brown, A.: The application of digital twin technology in operations and supply chain management: a bibliometric review. Supply Chain Manag. Int. J. **27**(2), 182–206 (2022)
20. Boschert, S.; Heinrich, C.; & Rosen, R.: Next generation digital twin. In Proc. Tmce, Vol. 2018, pp. 7–11. Las Palmas de Gran Canaria, Spain. (2018)
21. Botín-Sanabria, D.M.; Mihaita, A.S.; Peimbert-García, R.E.; Ramírez-Moreno, M.A.; Ramírez-Mendoza, R.A.; Lozoya-Santos, J.D.J.: Digital twin technology challenges and applications: a comprehensive review. Remote Sens. **14**(6), 1335 (2022)
22. Chaudhary, G.; Khari, M.; Elhoseny, M. (eds.): Digital Twin Technology. CRC Press, Boca Raton (2021)
23. Evangeline, P.: Digital twin technology for "smart manufacturing". In: Advances in Computers, Vol. 117, No. 1, pp. 35–49. Elsevier (2020)
24. Fan, C.; Jiang, Y.; Mostafavi, A.: Social sensing in disaster city digital twin: Integrated textual–visual–geo framework for situational awareness during built environment disruptions. J. Manag. Eng. **36**(3), 04020002 (2020)
25. Xie, Y.; Ebad Sichani, M.; Padgett, J.E.; DesRoches, R.: The promise of implementing machine learning in earthquake engineering: a state-of-the-art review. Earthq. Spectra **36**(4), 1769–1801 (2020)
26. Ortiz, B.; Kahn, L.; Bosch, M.; Bogden, P.; Pavon-Harr, V.; Savas, O.; & McCulloh, I.: Improving community resiliency and emergency response with artificial intelligence. arXiv preprint arXiv:2005.14212 (2020)
27. Habib, A.; Youssefi, I.; Kunt, M.M.: Identification of pulse-like ground motions using artificial neural network. Earthq. Eng. Eng. Vib. **21**(4), 899–912 (2022)
28. Du, A.; Wang, X.; Xie, Y.; Dong, Y.: Regional seismic risk and resilience assessment: methodological development, applicability, and future research needs—an earthquake engineering perspective. Reliab. Eng. Syst. Saf. **233**, 109104 (2023)
29. Habib, M.; Okayli, M.: Evaluating the sensitivity of machine learning models to data preprocessing technique in concrete compressive strength estimation. Arab. J. Sci. Eng. (2024). https://doi.org/10.1007/s13369-024-08776-2
30. Habib, M.; Bashir, B.; Alsalman, A.; Bachir, H.: Evaluating the accuracy and effectiveness of machine learning methods for

rapidly determining the safety factor of road embankments. Multidiscip. Model. Mater. Struct. **19**(5), 966–983 (2023)
31. Al-Houri, A.; Habib, A.; Al-Sadoon, Z.A.: Artificial intelligence-based design and analysis of passive control structures: an overview. J. Soft Comput. Civ. Eng. (2024). https://doi.org/10.22115/scce.2024.450722.1832
32. Albahri, A.S.; Khaleel, Y.L.; Habeeb, M.A.; Ismael, R.D.; Hameed, Q.A.; Deveci, M.; Alzubaidi, L.: A systematic review of trustworthy artificial intelligence applications in natural disasters. Comput. Electr. Eng. **118**, 109409 (2024)
33. Habib, A.; Barakat, S.; Al-Toubat, S.; Junaid, M.T.; Maalej, M.: Developing machine learning models for identifying the failure potential of fire-exposed FRP-strengthened concrete beams. Arab. J. Sci. Eng. (2024). https://doi.org/10.1007/s13369-024-09497-2
34. Freddi, F.; Galasso, C.; Cremen, G.; Dall'Asta, A.; Di Sarno, L.; Giaralis, A.; Woo, G.: Innovations in earthquake risk reduction for resilience: recent advances and challenges. Int. J. Disaster Risk Reduct **60**, 102267 (2021)
35. Habib, A.; Yildirim, U.: Influence of isolator properties and earthquake characteristics on the seismic behavior of RC structure equipped with quintuple friction pendulum bearings. Int. J. Struct. Stab. Dyn. **23**(06), 2350060 (2023)
36. Menna, C.; Felicioni, L.; Negro, P.; Lupíšek, A.; Romano, E.; Prota, A.; Hájek, P.: Review of methods for the combined assessment of seismic resilience and energy efficiency towards sustainable retrofitting of existing European buildings. Sustain. Cities Soc. **77**, 103556 (2022)
37. Jena, R.; Shanableh, A.; Al-Ruzouq, R.; Pradhan, B.; Gibril, M.B.A.; Khalil, M.A.; Ghamisi, P.: Explainable artificial intelligence (XAI) model for earthquake spatial probability assessment in Arabian peninsula. Remote Sens. **15**(9), 2248 (2023)
38. Habib, A.; Yildirim, U.: Distribution of strong input energy in base-isolated structures with complex nonlinearity: a parametric assessment. Multidiscip. Model. Mater. Struct. **19**(2), 324–340 (2023)
39. Li, Z.; Lim, H.W.; Li, N.; Long, Y.; Fang, D.: Assessing the seismic resilience of a healthcare system: a hybrid modeling. Int. J. Disaster Risk Reduct. **93**, 103730 (2023)
40. Habib, A.; Yildirim, U.; Eren, O.: Seismic behavior and damping efficiency of reinforced rubberized concrete jacketing. Arab. J. Sci. Eng. **46**(5), 4825–4839 (2021)
41. Xu, G.; Guo, T.; Li, A.; Zhang, H.; Wang, K.; Xu, J.; Dang, L.: Seismic resilience enhancement for building structures: a comprehensive review and outlook. Structures **59**, 105738 (2024)
42. Farsi, M.; Daneshkhah, A.; Hosseinian-Far, A.; Jahankhani, H. (eds.): Digital twin technologies and smart cities. Springer, Berlin (2020)
43. Ford, D.N.; Wolf, C.M.: Smart cities with digital twin systems for disaster management. J. Manag. Eng. **36**(4), 04020027 (2020)
44. Fuller, A.; Fan, Z.; Day, C.; Barlow, C.: Digital twin: enabling technologies, challenges and open research. IEEE Access **8**, 108952–108971 (2020)
45. Hayat, P.: Smart cities: a global perspective. India Q. **72**(2), 177–191 (2016)
46. He, B.; Bai, K.J.: Digital twin-based sustainable intelligent manufacturing: a review. Adv. Manuf. **9**, 1–21 (2021)
47. He, R.; Chen, G.; Dong, C.; Sun, S.; Shen, X.: Data-driven digital twin technology for optimized control in process systems. ISA Trans. **95**, 221–234 (2019)
48. Jones, D.; Snider, C.; Nassehi, A.; Yon, J.; Hicks, B.: Characterising the Digital Twin: a systematic literature review. CIRP J. Manuf. Sci. Technol. **29**, 36–52 (2020)
49. Kaewunruen, S.; Xu, N.: Digital twin for sustainability evaluation of railway station buildings. Front. Built Environ. **4**, 77 (2018)
50. Kaewunruen, S.; Peng, S.; Phil-Ebosie, O.: Digital twin aided sustainability and vulnerability audit for subway stations. Sustainability **12**(19), 7873 (2020)
51. Kaewunruen, S.; Rungskunroch, P.; Welsh, J.: A digital-twin evaluation of net zero energy building for existing buildings. Sustainability **11**(1), 159 (2018)
52. Kaewunruen, S.; Sresakoolchai, J.; Ma, W.; Phil-Ebosie, O.: Digital twin aided vulnerability assessment and risk-based maintenance planning of bridge infrastructures exposed to extreme conditions. Sustainability **13**(4), 2051 (2021)
53. Kaewunruen, S.; Lian, Q.: Digital twin aided sustainability-based lifecycle management for railway turnout systems. J. Clean. Prod. **228**, 1537–1551 (2019)
54. Khajavi, S.H.; Motlagh, N.H.; Jaribion, A.; Werner, L.C.; Holmström, J.: Digital twin: vision, benefits, boundaries, and creation for buildings. IEEE Access **7**, 147406–147419 (2019)
55. Madni, A.M.; Madni, C.C.; Lucero, S.D.: Leveraging digital twin technology in model-based systems engineering. Systems **7**(1), 7 (2019)
56. Malykhina, G. F.; Tarkhov, D. A.; F Malykhina, G.; & A Tarkhov, D.: Digital twin technology as a basis of the industry in future. In: European Proceedings of Social and Behavioural Sciences, vol. 51 (2018)
57. Mashaly, M.: Connecting the twins: a review on digital twin technology & its networking requirements. Proc. Comput. Sci. **184**, 299–305 (2021)
58. Onan Demirel, H.; Irshad, L.; Ahmed, S.; Tumer, I.Y.: Digital twin-driven human-centered design frameworks for meeting sustainability objectives. J. Comput. Inf. Sci. Eng. **21**(3), 031012 (2021)
59. Park, J.; Yang, B.: GIS-enabled digital twin system for sustainable evaluation of carbon emissions: a case study of Jeonju city, south Korea. Sustainability **12**(21), 9186 (2020)
60. Pirlone, F.; Spadaro, I.; Candia, S.: More resilient cities to face higher risks. Case Genoa. Sustain. **12**(12), 4825 (2020)
61. Qi, Q.; Tao, F.; Hu, T.; Anwer, N.; Liu, A.; Wei, Y.; Nee, A.Y.C.: Enabling technologies and tools for digital twin. J. Manuf. Syst. **58**, 3–21 (2021)
62. Samarakkody, A.; Amaratunga, D.; Haigh, R.: Technological innovations for enhancing disaster resilience in smart cities: a comprehensive urban scholar's analysis. Sustainability **15**(15), 12036 (2023)
63. Shah, S.A.; Seker, D.Z.; Rathore, M.M.; Hameed, S.; Yahia, S.B.; Draheim, D.: Towards disaster resilient smart cities: Can internet of things and big data analytics be the game changers? IEEE Access **7**, 91885–91903 (2019)
64. Tagliabue, L.C.; Cecconi, F.R.; Maltese, S.; Rinaldi, S.; Ciribini, A.L.C.; Flammini, A.: Leveraging digital twin for sustainability assessment of an educational building. Sustainability **13**(2), 480 (2021)
65. Tahmasebinia, F.; Lin, L.; Wu, S.; Kang, Y.; Sepasgozar, S.: Exploring the benefits and limitations of digital twin technology in building energy. Appl. Sci. **13**(15), 8814 (2023)
66. Tao, F.; Xiao, B.; Qi, Q.; Cheng, J.; Ji, P.: Digital twin modeling. J. Manuf. Syst. **64**, 372–389 (2022)
67. Tao, F.; Zhang, H.; Liu, A.; Nee, A.Y.: Digital twin in industry: state-of-the-art. IEEE Trans. Industr. Inf. **15**(4), 2405–2415 (2018)
68. VanDerHorn, E.; Mahadevan, S.: Digital Twin: generalization, characterization and implementation. Decis. Support Syst. **145**, 113524 (2021)
69. Wang, K.J.; Lee, T.L.; Hsu, Y.: Revolution on digital twin technology—a patent research approach. Int. J. Adv. Manuf. Technol. **107**, 4687–4704 (2020)

70. Yu, D.; He, Z.: Digital twin-driven intelligence disaster prevention and mitigation for infrastructure: advances, challenges, and opportunities. Nat. Hazards **112**(1), 1–36 (2022)
71. Yu, W.; Patros, P.; Young, B.; Klinac, E.; Walmsley, T.G.: Energy digital twin technology for industrial energy management: classification, challenges and future. Renew. Sustain. Energy Rev. **161**, 112407 (2022)
72. Zhang, X.; Shen, J.; Saini, P.K.; Lovati, M.; Han, M.; Huang, P.; Huang, Z.: Digital twin for accelerating sustainability in positive energy district: a review of simulation tools and applications. Front. Sustain. Cities **3**, 663269 (2021)
73. Zhu, S.; Li, D.; Feng, H.: Is smart city resilient? Evidence from China. Sustain. Cities Soc. **50**, 101636 (2019)
74. Zhu, S.; Li, D.; Feng, H.; Gu, T.; Hewage, K.; Sadiq, R.: Smart city and resilient city: differences and connections. Wiley Interdiscip. Rev. Data Min. Knowl. Discov. **10**(6), e1388 (2020)
75. Liu, C.; Chen, Y.; Xu, X.: Structural digital Twin for damage detection of CFRP composites using meta transfer learning-based approach. Expert Syst. Appl. **261**, 125458 (2025)
76. Pavón, R.M.; Alberti, M.G.; Álvarez, A.A.A.; Cepa, J.J.: Bim-based Digital Twin development for university Campus management. Case study ETSICCP. Expert Syst. Appl. **262**, 125696 (2025)
77. Cao, Y.; Li, S.; Shen, G.Q.; Chen, H.; Liu, Y.: Intelligent dynamic control of shield parameters using a hybrid algorithm and digital twin platform. Autom. Constr. **169**, 105882 (2025)
78. Jiang, F.; Li, J.; Ma, L.; Dong, Z.; Chen, W.; Broyd, T.; Wang, G.: Sustainable urban road planning under the digital twin-MCDM-GIS framework considering multidisciplinary factors. J. Clean. Prod. **469**, 143097 (2024)
79. Jiang, F.; Ma, L.; Broyd, T.; Chen, W.; Luo, H.: Digital twin enabled sustainable urban road planning. Sustain. Cities Soc. **78**, 103645 (2022)
80. Lehtola, V.V.; Koeva, M.; Elberink, S.O.; Raposo, P.; Virtanen, J.P.; Vahdatikhaki, F.; Borsci, S.: Digital twin of a city: review of technology serving city needs. Int. J. Appl. Earth Obs. Geoinf. **114**, 102915 (2022)
81. Bibri, S.E.; Huang, J.; Jagatheesaperumal, S.K.; Krogstie, J.: The synergistic interplay of artificial intelligence and digital twin in environmentally planning sustainable smart cities: a comprehensive systematic review. Environ. Sci. Ecotechnol. **20**, 100433 (2024)
82. Batty, M.: Digital twins in city planning. Nat. Comput. Sci. **4**(3), 192–199 (2024)
83. Caprari, G.; Castelli, G.; Montuori, M.; Camardelli, M.; Malvezzi, R.: Digital twin for urban planning in the green deal era: a state of the art and future perspectives. Sustainability **14**(10), 6263 (2022)
84. Elsehrawy, R.; Kumar, B.; Watson, R.: A digital twin uses classification system for urban planning & city infrastructure management. J. Inf. Technol. Constr. **26**, 832–862 (2021)
85. Charitonidou, M.: Urban scale digital twins in data-driven society: challenging digital universalism in urban planning decision-making. Int. J. Archit. Comput. **20**(2), 238–253 (2022)
86. Weil, C.; Bibri, S.E.; Longchamp, R.; Golay, F.; Alahi, A.: Urban Digital Twin Challenges: a systematic review and perspectives for sustainable smart cities. Sustain. Cities Soc. **99**, 104862 (2023)
87. Kliestik, T.; Poliak, M.; Popescu, G.H.: Digital twin simulation and modeling tools, computer vision algorithms, and urban sensing technologies in immersive 3D environments. Geopolit. History Int. Relat. **14**(1), 9–25 (2022)
88. Yang, S.; Kim, H.: Urban digital twin applications as a virtual platform of smart city. Int. J. Sustain. Build. Technol. Urban Dev. **12**(4), 363–379 (2021)
89. Shahat, E.; Hyun, C.T.; Yeom, C.: City digital twin potentials: a review and research agenda. Sustainability **13**(6), 3386 (2021)
90. White, G.; Zink, A.; Codecá, L.; Clarke, S.: A digital twin smart city for citizen feedback. Cities **110**, 103064 (2021)
91. Pelling, M.: The vulnerability of cities: natural disasters and social resilience. Routledge, London (2012)
92. Habib, M.; Habib, A.; Albzaie, M.; Farghal, A.: Sustainability benefits of AI-based engineering solutions for infrastructure resilience in arid regions against extreme rainfall events. Discov. Sustain. **5**(1), 278 (2024)
93. Hashem, I.A.T.; Chang, V.; Anuar, N.B.; Adewole, K.; Yaqoob, I.; Gani, A.; Chiroma, H.: The role of big data in smart city. Int. J. Inf. Manag. **36**(5), 748–758 (2016)
94. Rus, K.; Kilar, V.; Koren, D.: Resilience assessment of complex urban systems to natural disasters: a new literature review. Int. J. f Disaster Risk Reduct. **31**, 311–330 (2018)
95. Habib, A.; Yildirim, U.: Proposing unsupervised clustering-based earthquake records selection framework for computationally efficient nonlinear response history analysis of structures equipped with multi-stage friction pendulum bearings. Soil Dyn. Earthq. Eng. **182**, 108732 (2024)
96. Habib, A.; Junaid, M.T.; Dirar, S.; Barakat, S.; Al-Sadoon, Z.A.: Machine learning-based estimation of reinforced concrete columns stiffness modifiers for improved accuracy in linear response history analysis. J. Earthq. Eng. **29**, 1–26 (2024)
97. Toprak, S.; Wham, B.P.; Nacaroglu, E.; Ceylan, M.; Dal, O.; Senturk, A.E.: Impact of Seismic Geohazards on water supply systems and pipeline performance: insights from the 2023 Kahramanmaras Earthquakes. Eng. Geol. **340**, 107681 (2024)
98. Dinçer, A.E.; Dincer, N.N.; Tekin-Koru, A.; Yaşar, B.; Yılmaz, Z.: The impact of Kahramanmaraş (2023) earthquakes: a comparative case study for Adıyaman and Malatya. Int. J. Disaster Risk Reduct. **110**, 104647 (2024)
99. Samaei, S. R.: Using artificial intelligence to increase urban resilience: a case study of Tehran. In 13th International Conference on Advanced Research in Science, Engineering and Technology, Brussels, Belgium (2024)
100. Ray, P.P.; Mukherjee, M.; Shu, L.: Internet of things for disaster management: state-of-the-art and prospects. IEEE Access **5**, 18818–18835 (2017)
101. Khan, A.; Gupta, S.; Gupta, S.K.: Multi-hazard disaster studies: Monitoring, detection, recovery, and management, based on emerging technologies and optimal techniques. Int. J. Disaster Risk Reduct. **47**, 101642 (2020)
102. Esposito, M.; Palma, L.; Belli, A.; Sabbatini, L.; Pierleoni, P.: Recent advances in internet of things solutions for early warning systems: a review. Sensors **22**(6), 2124 (2022)
103. Habib, A.; Yildirim, U.: Modeling reinforced concrete moment frames supported on quintuple friction pendulum bearings for nonlinear response history analysis. J. Earthq. Tsunami **17**(02), 2350002 (2023)
104. Hasanuzzaman, M.; Hossain, S.; Shil, S.K.: Enhancing disaster management through AI-driven predictive analytics: improving preparedness and response. Int. J. Adv. Eng. Technol. Innov. **1**(01), 533–562 (2023)
105. Ling, M.; Thomas, R.J.: Data-driven disaster management: leveraging big data analytics for preparedness, response, and recovery. Int. J. Bus. Intell. Big Data Anal. **5**(1), 24–34 (2022)
106. Habib, A.; Houri, A.A.; Habib, M.; Elzokra, A.; Yildirim, U.: Structural performance and finite element modeling of roller compacted concrete dams: a review. Latin Am. J. Solids Struct. **18**(04), e376 (2021)
107. Wang, Z.: Digital twin technology. In: Industry 4.0-Impact on Intelligent Logistics and Manufacturing. IntechOpen (2020)
108. Erol, T.; Mendi, A. F.; & Doğan, D.: Digital transformation revolution with digital twin technology. In 2020 4th International

Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT), pp. 1–7. IEEE (2020)
109. Fan, C.; Zhang, C.; Yahja, A.; Mostafavi, A.: Disaster City Digital Twin: A vision for integrating artificial and human intelligence for disaster management. Int. J. Inf. Manage. **56**, 102049 (2021)

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